Secant–hyperplane defects of complete caps

Tavis Rudd

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Abstract

A cap in $\mathrm{PG}(d,q)$ is complete when every point off it lies on a secant. We show that the overlap loss of a complete k-cap, the number $\binom{k}{2}(q-1)-\theta_d+k$ of coincidences among the points its secants cover, is carried by every hyperplane in a prescribed amount: the loss inside a hyperplane meeting the cap in s points is the quadratic $\binom{k}{2}-\theta_{d-1}+q\binom{s}{2}-(k-2)s$. Since that quantity lies between zero and the total loss, the hyperplane sections of a complete cap have sizes in a finite admissible set, which near the counting bound is very small. Feeding the admissible set into the hyperplane character equations, globally and through a fixed secant, gives integer feasibility conditions and a lower bound on the number of coplanar quadruples through every secant. These exclude complete caps of size 31 in $\mathrm{PG}(4,7)$, 37 in $\mathrm{PG}(4,8)$, 97 in $\mathrm{PG}(4,16)$, 293 in $\mathrm{PG}(6,8)$, and 387 in $\mathrm{PG}(6,9)$, each equal to the counting bound. A formal model shows that the degree constraints on coplanar quadruples cannot by themselves force collisions on a positive proportion of secants, so the hyperplane constraint is an essential ingredient. The identity extends to caps complete outside a prescribed set of holes.

Keywords. complete cap; hyperplane section; quasi-perfect code; covering radius; character equations; coplanar quadruple; free pair.

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1 Introduction

A k-cap in the projective space $\mathrm{PG}(d,q)$ over the field of order q is a set of k points, no three collinear. A line meeting the cap in two points is a secant, and the cap is complete if every point off it lies on a secant, that is, if it is maximal under inclusion. The smallest size of a complete cap in $\mathrm{PG}(d,q)$ is denoted $t_2(d,q)$. Counting the points that the $\binom{k}{2}$ secants can cover gives the classical lower bound

$$\binom{k}{2}(q-1)+k \geq \theta_d, \qquad \theta_j=\frac{q^{j+1}-1}{q-1}, \tag{1.1}$$

so $t_2(d,q) \geq \sqrt{2}\,q^{(d-1)/2}$ up to lower-order terms. In the plane this bound has been improved by Segre's lemma of tangents [12, 9]; in $\mathrm{PG}(3,q)$ a secant-local count improves the additive constant [11]; for $d \geq 4$ the literature quotes (1.1) as the lower bound, and the exact value of $t_2(d,q)$ for $d \geq 3$ is known only for very small q [3, 1, 2]. Complete caps are the same objects as parity-check matrices of quasi-perfect linear codes of minimum distance four and covering radius two, and $t_2(d,q)$ is the shortest length of such a code of codimension $d+1$ [3].

This paper adds one constraint to the counting argument and follows it to its finite consequences. The counting bound has a slack, the *overlap loss*

$$\Lambda_0=\binom{k}{2}(q-1)-\theta_d+k, \tag{1.2}$$

which counts, with multiplicity, the points covered by more than one secant. The observation is that this slack is not free to distribute: every hyperplane must carry a prescribed share of it.

Theorem 1.1 (Hyperplane loss identity)**.** *Let A be a k-cap in $\mathrm{PG}(d,q)$, $d \geq 2$, and for a point $x \notin A$ let $r(x)$ be the number of secants of A through x. For every hyperplane Π with $|A \cap \Pi| = s$,*

$$\sum_{x \in \Pi \setminus A} (r(x) - 1) = Q(s), \qquad Q(s) = \binom{k}{2} - \theta_{d-1} + q\binom{s}{2} - (k-2)s. \tag{1.3}$$

If A is complete then $0 \leq Q(s) \leq \Lambda_0$ for every hyperplane section size s.

The lower inequality is the counting bound (1.1) applied inside the hyperplane; the upper inequality says that a hyperplane cannot carry more loss than the whole space has. Together they confine the section sizes of a complete cap to the *admissible set*

$$\mathcal{S} = \{s : 0 \leq Q(s) \leq \Lambda_0\}. \tag{1.4}$$

Since Q is a quadratic in s with leading coefficient $q/2$, the admissible set consists of at most two short runs of integers around the roots of Q, and when Λ_0 is small, which is exactly the situation at the counting bound, it can have two elements. A complete 31-cap in $\mathrm{PG}(4,7)$, for instance, would have $\Lambda_0 = 20$ and would meet every hyperplane in 2 or 7 points.

Section sizes of a cap are tied to its parameters by the character equations, the three moment identities obtained by counting hyperplanes, point–hyperplane incidences, and pair–hyperplane incidences. Restricting those equations to the admissible set gives a finite integer system, and its infeasibility excludes the cap. The same equations restricted to the hyperplanes through a fixed secant ℓ bring in a second invariant, the number T_ℓ of coplanar four-subsets of A containing both endpoints of ℓ, which measures how many other secants ℓ meets.

Theorem 1.2 (Secant–hyperplane moments)**.** *Let A be a k-cap in $\mathrm{PG}(d,q)$, $d \geq 3$, let ℓ be a secant, and for each hyperplane $\Pi \supset \ell$ put $w_\Pi = |A \cap \Pi| - 2$. Then*

$$\#\{\Pi \supset \ell\} = \theta_{d-2}, \qquad \sum_{\Pi \supset \ell} w_\Pi = (k-2)\theta_{d-3}, \qquad \sum_{\Pi \supset \ell} \binom{w_\Pi}{2} = \binom{k-2}{2}\theta_{d-4} + q^{d-3}\, T_\ell. \tag{1.5}$$

For a complete cap the w_Π lie in $\mathcal{S} - 2$. Fixing the count and the first moment and minimizing the second over that support gives an unconditional lower bound on T_ℓ for every secant (Theorem 5.2), and the secant-local coverage bound of [11] converts a lower bound on every T_ℓ into a lower bound on Λ_0. Both routes close finitely.

Theorem 1.3 (Exclusions)**.** *There is no complete cap of size* 31 *in* $\mathrm{PG}(4,7)$, 37 *in* $\mathrm{PG}(4,8)$, 97 *in* $\mathrm{PG}(4,16)$, 293 *in* $\mathrm{PG}(6,8)$, *or* 387 *in* $\mathrm{PG}(6,9)$. *Hence*

$$t_2(4,7) \geq 32, \quad t_2(4,8) \geq 38, \quad t_2(4,16) \geq 98, \quad t_2(6,8) \geq 294, \quad t_2(6,9) \geq 388.$$

Each excluded size is the smallest k satisfying (1.1), so each bound improves the counting bound by one. In four of the five cases the admissible set has two elements and the global character equations are already inconsistent over it; the fifth, $\mathrm{PG}(6,8)$, has five admissible sizes separated by a gap of eleven around the mean, and is excluded by the coverage route.

A deterministic exact-arithmetic scan over $d \le 6$ and small q (Section 7) finds no other gain over the counting bound in dimension at least four, and never a gain of more than one. Section 8 explains why: one step above the counting bound Λ_0 grows by about kq, the admissible set becomes an interval, and every test is satisfied. The value of Theorem 1.3 is therefore the mechanism rather than the numbers, which remain far from the smallest known complete caps (56 points in $\mathrm{PG}(4,7)$ [1]).

The second contribution is negative. The companion paper [11] asks whether a positive proportion of the secants of a complete cap in $\mathrm{PG}(d,q)$, $d \ge 4$, must carry $T_\ell \ge cq$ collisions, which would improve the counting bound at the scale $q^{(d-3)/2}$. Proposition 8.1 constructs formal collision data satisfying every constraint that the coplanar-quadruple counts, the secant–point incidences, and the plane-pencil identities impose, in which only $o(\binom{k}{2})$ secants meet any other. Any proof of such a proportion must use an ambient constraint; Theorem 1.1 is one.

Related work

The global counting bound (1.1) and the dictionary between complete caps and quasi-perfect codes are standard; the papers of Bartoli, Davydov, Faina, Kreshchuk, Marcugini, and Pambianco cited above collect the known sizes and constructions of small complete caps and quote $\sqrt{2}\,q^{(d-1)/2}$ as the lower bound for $d \ge 3$. The character equations for hyperplane sections are classical; in coding terms they are the first two power moment identities of Pless [8] for the projective code generated by the cap, and Theorem 1.2 is the same identity for the code shortened at two coordinates. Counting cap points over the hyperplanes through a fixed subspace is a standard device, used for instance by Thas over the hyperplanes through a plane to bound the largest caps of $\mathrm{PG}(n,q)$, q even [13]. Divisibility conditions on hyperplane intersection numbers are the subject of the theory of divisible codes [6]; the divisibility used here is imposed by completeness rather than assumed. Wehlau classifies the complete caps of $\mathrm{PG}(n,2)$ that miss a codimension-two subspace through their hyperplane intersection sizes [14], the nearest instance we know of constraining a complete cap by its hyperplane sections; the mechanism there is the binary structure, not a loss count. Pairs with $T_\ell = 0$ are the free pairs of Farr and Lisoněk [4, 5], and caps in which every pair is free are the 4-general sets of Pavese [7]; both notions reappear in Section 8. The overlap loss Λ_0 and the secant-local statistics T_ℓ and φ_m are taken from [11], and the integer-envelope viewpoint from [10]; the present paper differs from both in that its degree sequences are hyperplane section sizes rather than secant occupancies or point–block incidences. The lower half of Theorem 1.1 is a one-line consequence of completeness that we expect to be folklore, though we did not find it stated; we found no earlier statement of the upper half, of the admissible set, or of any exclusion of a counting-bound size for a complete cap in $\mathrm{PG}(d,q)$ with $d \ge 4$. The searches and sources behind that sentence are recorded with the paper's repository.

Organization

Section 2 fixes notation and recalls the pencil and character identities. Sections 3 and 4 prove Theorems 1.1 and 1.2, including the version for caps complete outside a prescribed set of holes. Section 5 derives the integer feasibility conditions and the local bounds on T_ℓ, Section 6 converts them into coverage bounds, and Section 7 proves Theorem 1.3 and describes the scan. Section 8 gives the formal countermodel and the reasons the method stops one step above the counting bound.

2 Caps, secants, and two families of identities

Throughout, A is a k-cap in $\mathrm{PG}(d,q)$ with $k \geq 4$, and $d \geq 3$ except where $d \geq 2$ is stated. We write $\theta_j = (q^{j+1}-1)/(q-1)$ for the number of points of $\mathrm{PG}(j,q)$, with $\theta_j = 0$ for $j < 0$, and put

$$N = \binom{k}{2}, \qquad m = \lfloor k/2 \rfloor.$$

A *secant* is a line meeting A in two points; there are N of them, and each carries $q-1$ points off A. For $x \notin A$ the *index* $r(x)$ is the number of secants through x; the *covered* points are those with $r(x) \geq 1$, and A is complete exactly when every point off A is covered. Two secants through a common point $x \notin A$ have four distinct endpoints, since three cap points are never collinear, so $r(x) \leq m$.

Counting secant–point incidences,

$$\sum_{x \notin A} r(x) = N(q-1), \tag{2.1}$$

and writing Y for the set of covered points, the *overlap loss*

$$\Lambda(A) = \sum_{x \in Y} (r(x)-1) = N(q-1) - |Y| \tag{2.2}$$

is nonnegative, equals $\Lambda_0 = N(q-1) - \theta_d + k$ when A is complete, and exceeds Λ_0 otherwise, by exactly the number of uncovered points. The counting bound (1.1) is the statement $\Lambda_0 \geq 0$.

2.1 Coplanar quadruples through a secant

Let $c_4(A)$ be the number of four-subsets of A contained in a plane. For a secant ℓ with endpoints a, b put

$$T_\ell = \#\{B \subseteq A : |B| = 4,\ a, b \in B,\ B \text{ coplanar}\}.$$

The θ_{d-2} planes through ℓ partition $A \setminus \{a, b\}$; if the plane π contains z_π of these points then $\{a, b, c, d\}$ is coplanar exactly when c and d lie in the same plane through ℓ, so

$$T_\ell = \sum_{\pi \supset \ell} \binom{z_\pi}{2}, \qquad \sum_{\pi \supset \ell} z_\pi = k-2, \qquad \sum_\ell T_\ell = 6\, c_4(A). \tag{2.3}$$

Two secants with disjoint endpoints meet if and only if their four endpoints are coplanar, and then they meet at a point off A. Hence T_ℓ also counts the secants meeting ℓ outside A, and a point $x \in \ell \setminus A$ accounts for $r(x)-1$ of them:

$$T_\ell = \sum_{x \in \ell \setminus A} (r(x)-1). \tag{2.4}$$

These identities are from [11, Section 5], where a lower bound on every T_ℓ is converted into a lower bound on $\Lambda(A)$; we recall that conversion in Section 6. A pair $\{a, b\}$ with $T_{ab} = 0$ is one such that every plane through a and b contains at most one further cap point, a *free pair* in the sense of Farr and Lisoněk [4, 5].

2.2 Character equations

For a hyperplane Π write $s_\Pi = |A \cap \Pi|$. Counting the hyperplanes through no point, one point, and two points of A gives the classical character equations

$$\sum_{\Pi} 1 = \theta_d, \qquad \sum_{\Pi} s_\Pi = k\,\theta_{d-1}, \qquad \sum_{\Pi} \binom{s_\Pi}{2} = N\,\theta_{d-2}, \tag{2.5}$$

the sums running over all hyperplanes; the third uses that two points span a line, which lies in θ_{d-2} hyperplanes. These hold for every point set and are the standard starting point for the study of intersection numbers.

2.3 The coding dictionary

A complete cap spans $\mathrm{PG}(d,q)$, since a cap inside a hyperplane covers no point outside it. Take a generator matrix G with the k points of A as columns and let C_A be its row space, a projective $[k, d+1]_q$ code. Its codewords are the linear functionals restricted to A, so they correspond to hyperplanes, and the codeword attached to Π has weight $k - s_\Pi$. The dual code $C_A^\perp$ has minimum distance at least four because no three columns are dependent, and its weight-four words are the dependencies among coplanar quadruples: each quadruple contributes $q-1$ words with all four coefficients nonzero, and conversely. Completeness of A says that every vector of $\mathbb{F}_q^{d+1}$ is a combination of at most two columns of G, that is, that $C_A^\perp$ has covering radius two; a linear code with minimum distance four and covering radius two is quasi-perfect. Thus complete k-caps in $\mathrm{PG}(d,q)$ are exactly the parity-check matrices of quasi-perfect $[k, k-d-1, 4]_q$ codes, with the two exceptions of the 5-cap in $\mathrm{PG}(3,2)$ and the 11-cap in $\mathrm{PG}(4,3)$, whose codes are perfect with minimum distance five [3]. In this dictionary Λ_0 is the excess of the sphere-covering count for radius two, the admissible set (1.4) is a restriction on the weight distribution of C_A, and Theorem 1.2 concerns the weight distribution of C_A shortened at the two coordinates of a secant.

3 The hyperplane loss identity

The identity is a count of secant–point incidences inside a hyperplane, with the cap points included at a fictitious multiplicity so that every secant contributes the same way.

Proof of Theorem 1.1. Extend the index to the cap points by $r(a) = k-1$ for $a \in A$, the number of secants through a. Every secant meets Π in exactly one point unless it lies in Π, in which case it contributes $q+1$ incidences, and the secants inside Π are the $\binom{s}{2}$ pairs of $A \cap \Pi$. Hence

$$\sum_{x\in\Pi} r(x) = N + q\binom{s}{2}.$$

Subtracting one for each of the θ_{d-1} points of Π, and then $k-2$ for each of the s cap points, leaves $\sum_{x\in\Pi\setminus A}(r(x)-1) = N + q\binom{s}{2} - \theta_{d-1} - (k-2)s$, which is (1.3). If A is complete then every summand $r(x)-1$ is nonnegative, so $Q(s) \geq 0$, and the summands over all of $\mathrm{PG}(d,q) \setminus A$ add to Λ_0 by (2.2), so $Q(s) \leq \Lambda_0$. $\square$

Summing (1.3) over all hyperplanes and using the character equations (2.5) returns $\theta_{d-1}\sum_{x\notin A}(r(x)-1) = \theta_{d-1}\Lambda(A)$ on both sides, as it must; the content of the theorem is the distribution, not the total. For $d=2$ the hyperplanes are lines, $s \leq 2$, and the identity reads $Q(0) = N-q-1$, $Q(1) = N-k-q+1$, $Q(2) = N-2k+q+3$: a complete arc in $\mathrm{PG}(2,q)$ has $N \geq q+1$ with the loss on an external line equal to $N-q-1$. The planar case carries no new information. For $d \geq 3$ the quadratic Q has an interior minimum near $s = (k-2)/q + \frac{1}{2}$, and the shape of the admissible set is governed by two numbers.

Lemma 3.1 (Shape of the admissible set). *Let A be a complete k-cap in $\mathrm{PG}(d, q)$, $d \geq 3$. If Q has real roots $\sigma_1 \leq \sigma_2$, then every hyperplane section size lies in*

$$\mathcal{S} = \{s \leq \sigma_1 : Q(s) \leq \Lambda_0\} \cup \{s \geq \sigma_2 : Q(s) \leq \Lambda_0\},$$

two runs of consecutive integers, each with at most $2+\sqrt{2\Lambda_0/q}$ elements; no section size lies strictly between the roots. If Q has no real root, then $\mathcal{S}$ is the single run $\{s : Q(s) \leq \Lambda_0\}$.

Proof. The vertex of Q is at $s^* = \frac{1}{2}+(k-2)/q$, and on each side of it Q is monotone, so each run is an interval of integers. For $s \geq s^*$ and $j \geq 0$, $Q(s+j)-Q(s) = j\big(qs-(k-2)\big)+q\binom{j}{2} \geq q\binom{j}{2}$, and for $s \leq s^*$, $Q(s-j) - Q(s) = j\big((k-2) - qs\big) + q\binom{j+1}{2} \geq q\binom{j}{2}$. If both endpoints of a run are admissible, the difference of their Q-values is at most Λ_0, so $q\binom{j}{2} \leq \Lambda_0$ for the distance j between them. □

At the counting bound $\Lambda_0 < k(q-1)+1$, so each run has $O(\sqrt{k})$ elements, while the two runs are separated by the distance between the roots of Q. For the five parameter sets of Theorem 1.3 the runs have at most three elements and the gap between them is at least four; Table 1 lists them.

3.1 Prescribed holes

The identity is exact for every cap, so it can be applied when only part of the space must be covered. Let $\mathcal{H}$ be a set of h points off A, and call A complete outside $\mathcal{H}$ if every point outside $A \cup \mathcal{H}$ is covered; the notion is studied in [11], where $\mathcal{H}$ is a conic, a quadric, or an arbitrary set.

Proposition 3.2 (Loss identity with holes). *Let A be a k-cap complete outside $\mathcal{H}$, $|\mathcal{H}| = h$, and let $u \leq h$ be the number of uncovered points. Then for every hyperplane Π,*

$$-|\mathcal{H} \cap \Pi| \leq -\#\{x \in \mathcal{H} \cap \Pi : r(x) = 0\} \leq Q(s_\Pi) \leq \Lambda_0 + u \leq \Lambda_0 + h.$$

In particular every section size lies in $\{s : -h \leq Q(s) \leq \Lambda_0 + h\}$.

Proof. In (1.3) the summands $r(x) - 1$ are at least -1, with equality exactly at uncovered points, all of which lie in $\mathcal{H}$; this gives the lower bounds. The nonnegative summands over the whole space add to $\Lambda(A)+u$ with $\Lambda(A)$ as in (2.2), and $\Lambda(A) = N(q-1)-(\theta_d-k-u) = \Lambda_0 + u$. □

Every statement below that uses only the admissible set and the moment identities holds for caps complete outside $\mathcal{H}$ after replacing Λ_0 by $\Lambda_0 + h$ and $\mathcal{S}$ by the wider set of the proposition. We do not pursue the numerical consequences here.

4 Secant–hyperplane moments

The character equations (2.5) count hyperplanes through zero, one, and two cap points. Restricting them to the hyperplanes through a fixed secant leaves the same three counts, but the pairs of further cap points now split according to whether they are coplanar with the secant, and that is where T_ℓ enters.

Proof of Theorem 1.2. Fix the secant $\ell = ab$. The hyperplanes through ℓ correspond to the hyperplanes of the quotient space $\mathrm{PG}(d-2,q)$, so there are θ_{d-2} of them. A further cap point u spans with ℓ a plane, which lies in θ_{d-3} hyperplanes; summing over the $k-2$ choices of u gives the first moment. For a pair $\{u,v\} \subseteq A \setminus \{a,b\}$, the span of ℓ, u, v is a plane when $\{a,b,u,v\}$ is coplanar and a solid otherwise, since no three of the four points are collinear. A plane lies in θ_{d-3} hyperplanes and a solid in θ_{d-4}; there are T_ℓ coplanar pairs and $\binom{k-2}{2} - T_\ell$ others, so

$$\sum_{\Pi \supset \ell} \binom{w_\Pi}{2} = \Big(\binom{k-2}{2} - T_\ell\Big)\theta_{d-4} + T_\ell\,\theta_{d-3} = \binom{k-2}{2}\theta_{d-4} + q^{d-3}\,T_\ell,$$

because $\theta_{d-3} - \theta_{d-4} = q^{d-3}$. $\square$

For $d = 3$ the hyperplanes through ℓ are its planes, $\theta_{-1} = 0$, and the second moment is T_ℓ itself, which is the pencil identity (2.3). For $d \geq 4$ the theorem expresses T_ℓ through the section sizes of the hyperplanes through ℓ,

$$T_\ell = \frac{1}{q^{d-3}}\Big(\sum_{\Pi \supset \ell} \binom{w_\Pi}{2} - \binom{k-2}{2}\theta_{d-4}\Big), \tag{4.1}$$

so the sum on the right is congruent to $\binom{k-2}{2}\theta_{d-4}$ modulo q^{d-3} for every secant.

Remark 4.1 (Shortened codes)*.* In the dictionary of Section 2, the hyperplanes through $\ell = ab$ are the codewords of C_A vanishing at the coordinates a and b, that is, the code C_A shortened at $\{a,b\}$, a $[k-2, d-1]_q$ code whose codewords have weights $k - 2 - w_\Pi$. Its dual is $C_A^\perp$ punctured at $\{a,b\}$, whose words of weight at most two are the $(q-1)T_\ell$ images of the weight-four words of $C_A^\perp$ with support containing a and b. Theorem 1.2 is therefore the first two power moment identities of Pless [8] for the shortened code, which express its zeroth, first, and second weight moments through its length, dimension, and the numbers of dual words of weight one and two. We keep the geometric proof because it exhibits the plane-versus-solid dichotomy that produces the factor q^{d-3}.

The same restriction can be made to the hyperplanes through any fixed subspace; counting over the hyperplanes through a fixed plane is the technique of Thas [13] for bounding the largest caps of $\mathrm{PG}(n,q)$, q even. The secant is the right subspace here because the second moment then measures the collisions of one secant with the others, the quantity that the coverage bound of Section 6 consumes.

5 Integer feasibility and local bounds

For a complete cap the section sizes lie in the admissible set $\mathcal{S}$ of (1.4). Writing b_s for the number of hyperplanes with section size s, the character equations (2.5) become a system of three linear equations in the unknowns $(b_s)_{s\in\mathcal{S}}$, which must have a solution in nonnegative integers.

Proposition 5.1 (Global feasibility)**.** *If A is a complete k-cap in $\mathrm{PG}(d,q)$, then there are nonnegative integers $(b_s)_{s\in\mathcal{S}}$ with*

$$\sum_{s\in\mathcal{S}} b_s = \theta_d, \qquad \sum_{s\in\mathcal{S}} s\,b_s = k\,\theta_{d-1}, \qquad \sum_{s\in\mathcal{S}} \binom{s}{2} b_s = N\,\theta_{d-2}.$$

When $\mathcal{S} = \{s_1, s_2\}$ has two elements the first two equations determine $b_{s_2} = (k\theta_{d-1} - s_1\theta_d)/(s_2 - s_1)$, which must be an integer between 0 and θ_d, and the third equation must then hold as well. This is the test that settles four of the five cases of Theorem 1.3.

The local version, through a fixed secant, is the same system with the counts of Theorem 1.2; its second moment is not fixed but is tied to T_ℓ. Put

$$t = \theta_{d-2}, \qquad R = (k-2)\theta_{d-3}, \qquad C_0 = \binom{k-2}{2}\theta_{d-4}, \qquad D = q^{d-3}, \tag{5.1}$$

and let $\mathcal{S} - 2 = \{s - 2 : s \in \mathcal{S},\ s \geq 2\}$ be the admissible values of w_Π.

Theorem 5.2 (Local bounds)**.** *Let A be a complete k-cap in* $\mathrm{PG}(d, q)$*, $d \geq 3$, and let ℓ be a secant. Then there are nonnegative integers $(b_w)_{w \in \mathcal{S}-2}$, the numbers of hyperplanes through ℓ of each section size, with*

$$\sum_w b_w = t, \qquad \sum_w w\, b_w = R, \qquad \sum_w \binom{w}{2} b_w = C_0 + D\, T_\ell. \tag{5.2}$$

Consequently, if $P_{\min}$ and $P_{\max}$ are the smallest and largest values of $\sum_w \binom{w}{2} b_w$ over the nonnegative integer solutions of the first two equations that satisfy $\sum_w \binom{w}{2} b_w \equiv C_0 \pmod{D}$, then

$$\max\Big\{0,\ \Big\lceil \frac{P_{\min} - C_0}{D} \Big\rceil\Big\} \leq T_\ell \leq \min\Big\{(q-1)(m-1),\ \Big\lfloor \frac{P_{\max} - C_0}{D} \Big\rfloor\Big\}, \tag{5.3}$$

and if the first two equations have no nonnegative integer solution, or no solution meets the congruence, then no such cap exists.

Proof. The three equations are Theorem 1.2 grouped by section size, and the support is $\mathcal{S} - 2$ by Theorem 1.1. The bounds in (5.3) follow, the trivial upper bound $(q-1)(m-1)$ coming from (2.4) and $r(x) \leq m$. □

The extrema $P_{\min}$ and $P_{\max}$ are the values of an integer program in at most $|\mathcal{S}|$ variables; for $|\mathcal{S}| \leq 3$ the solutions are enumerated directly, and in general a closed-form lower bound is available whenever the mean R/t falls into a gap of the support.

Lemma 5.3 (Bracket bound)**.** *Let W be a finite set of nonnegative integers and let $(b_w)_{w \in W}$ be nonnegative integers with $\sum b_w = t$ and $\sum w\, b_w = R$. Let $a \leq b$ be elements of W with $a \leq R/t \leq b$ and no element of W strictly between a and b. Then*

$$\sum_{w \in W} \binom{w}{2} b_w = \frac{(a+b-1)R - abt}{2} + \frac{1}{2} \sum_{w \in W} (w-a)(w-b)\, b_w, \tag{5.4}$$

where every summand of the last sum is nonnegative, and a summand with $w \notin \{a, b\}$ is at least $(b - a + 1)\, b_w$.

Proof. For every integer w, $\binom{w}{2} = \frac{1}{2}\big((a+b-1)w - ab\big) + \frac{1}{2}(w-a)(w-b)$; multiply by b_w and sum. Since W has no element strictly between a and b, every $w \in W$ satisfies $w \leq a$ or $w \geq b$, so $(w-a)(w-b) \geq 0$, and if $w \neq a, b$ then one factor is at least 1 and the other at least $b - a + 1$. □

Corollary 5.4 (Closed-form local bound)**.** *In the setting of Theorem 5.2, let $a \leq b$ bracket R/t in $\mathcal{S} - 2$ as in Lemma 5.3 and put*

$$\tau_0 = \frac{(a+b-1)R - abt - 2C_0}{2D}.$$

Then $T_\ell \geq \lceil \tau_0 \rceil$ for every secant, and the number of hyperplanes through ℓ whose section size is not $a+2$ or $b+2$ is at most $2D(T_\ell - \tau_0)/(b - a + 1)$.

Proof. Apply Lemma 5.3 to (5.2): $2D(T_\ell - \tau_0) = \sum_w (w-a)(w-b)b_w \geq 0$, and the last sum is at least $(b-a+1)$ times the number of hyperplanes with $w \notin \{a,b\}$. $\square$

The second statement is a stability estimate: a secant whose collision count is close to τ_0 is contained almost only in hyperplanes of the two bracketing sizes. When R/t is itself admissible the bracket degenerates to $a = b$ and $\tau_0 = (t\binom{a}{2} - C_0)/D$, the balanced case. The lower bound of Corollary 5.4 is what a cap must exhibit on *every* secant; it is compared in Section 6 with what the total loss can afford.

Remark 5.5 (One-sided version). Both Theorem 5.2 and Corollary 5.4 remain valid with $\mathcal{S}$ replaced by the larger set $\mathcal{S}_+ = \{s : Q(s) \geq 0\}$, which uses only the lower half of Theorem 1.1. The bracket then spans the gap between the roots of Q alone. For the five cases of Theorem 1.3 this weaker support still gives a positive τ_0; see Table 1.

6 From collision counts to coverage

A lower bound on every T_ℓ is a lower bound on how many secants each secant meets. Each meeting costs coverage, and the cost is smallest when the meetings on a secant are concentrated at few points of high index. The exact minimum cost is the following envelope from [11, Section 5]; we include the short proof.

For $m \geq 2$ and an integer $T \geq 0$ write $T = j(m-1) + \rho$ with $0 \leq \rho < m-1$ and put

$$\varphi_m(T) = j\,\frac{m-1}{m} + \frac{\rho}{\rho+1}. \tag{6.1}$$

Thus $\varphi_m(T) = T/(T+1)$ for $T \leq m-1$, and φ_m is nondecreasing with $\varphi_m(T) \geq \max\{T/m,\, T/(T+1)\}$.

Theorem 6.1 (Secant-local coverage bound)**.** *For every k-cap A in* $\mathrm{PG}(d,q)$*,*

$$\Lambda(A) \geq \sum_\ell \varphi_m(T_\ell),$$

the sum running over the secants of A. In particular a complete cap on all of whose secants $T_\ell \geq \beta$ satisfies $\Lambda_0 \geq N\varphi_m(\beta)$.

Proof. Distribute the loss over the secants: a covered point of index r gives $(r-1)/r$ to each of its r secants, so $\Lambda(A) = \sum_\ell L_\ell$ with $L_\ell = \sum_{x \in \ell \setminus A}(1 - 1/r(x))$. On a fixed secant the indices are integers in $[1,m]$ with $\sum(r(x)-1) = T_\ell$ by (2.4). The function $r \mapsto 1 - 1/r$ is concave, so the minimum of L_ℓ over real vectors with these constraints is attained at a vertex of the constraint polytope, where at most one coordinate lies strictly between 1 and m. Coordinates equal to 1 contribute nothing, j coordinates equal to m contribute $j(m-1)/m$ and use $j(m-1)$ of the total, and the remaining coordinate $\rho+1$ contributes $\rho/(\rho+1)$; so $L_\ell \geq \varphi_m(T_\ell)$. The last statement uses monotonicity of φ_m and $\Lambda(A) = \Lambda_0$ for a complete cap. $\square$

Combined with Corollary 5.4 this gives the second exclusion test: a complete cap must satisfy

$$N\varphi_m(\lceil \tau_0 \rceil) \leq \Lambda_0. \tag{6.2}$$

Near the counting bound the left side is of order N as soon as $\tau_0 > 0$, while the right side is at most $k(q-1)$, so the test is decisive whenever the admissible support has a gap around the mean. A gap is the typical situation only at the counting bound itself; see Section 8.

The collisions on one secant are also bounded above by the total loss, because the planes through the secant carry planar arcs whose own losses add. The plane-fan bound of [11, Section 5] states that every secant of a k-cap in $\mathrm{PG}(d,q)$, $d \geq 3$, satisfies $\Lambda(A) \geq 2T_\ell + 2T_\ell^2/(k-2)$, so for a complete cap

$$T_\ell \leq U := \left\lfloor \tfrac{1}{2}\left(\sqrt{(k-2)^2 + 2(k-2)\Lambda_0} - (k-2)\right)\right\rfloor. \tag{6.3}$$

A lower bound $\lceil \tau_0 \rceil > U$ would be a third exclusion test. In the scan of Section 7 it never fires where the first two do not, and we record it only for completeness of the certificate.

7 Exclusions at the counting bound

We first work one case by hand, then give the data for all five, and finally describe the exact scan that found them.

7.1 A complete 31-cap in $\mathrm{PG}(4,7)$

Suppose A is a complete 31-cap in $\mathrm{PG}(4,7)$. Then $N = 465$, $\theta_3 = 400$, $\theta_4 = 2801$, and $\Lambda_0 = 465 \cdot 6 - 2801 + 31 = 20$; the counting bound is met with a slack of 20, so 31 is its smallest admissible value. The hyperplane loss polynomial is

$$Q(s) = 465 - 400 + 7\binom{s}{2} - 29s = \tfrac{1}{2}(7s^2 - 65s + 130),$$

with $Q(0) = 65$, $Q(1) = 36$, $Q(2) = 14$, $Q(3) = -1$, $Q(6) = -4$, $Q(7) = 9$, $Q(8) = 29$, and Q is decreasing before its vertex at $s^* = 4.64$ and increasing after. The only values in $[0,20]$ are $Q(2) = 14$ and $Q(7) = 9$: every solid meets A in exactly 2 or exactly 7 points.

Global route. With b_2 solids of size 2 and b_7 of size 7, the first two character equations read $b_2 + b_7 = 2801$ and $2b_2 + 7b_7 = 31 \cdot 400 = 12400$, so $5b_7 = 6798$, which is not divisible by 5. No such cap exists.

Local route. Fix a secant ℓ. The $t = \theta_2 = 57$ solids through ℓ have $w_\Pi \in \{0,5\}$ with $\sum w_\Pi = R = 29 \cdot 8 = 232$, and 232 is not divisible by 5; the local system is infeasible as well. Using only the lower half $Q(s) \geq 0$, the section sizes lie in $\{0,1,2\} \cup \{7,\ldots,31\}$, so $w_\Pi \in \{0\} \cup \{5,\ldots,29\}$ and the mean $R/t = 4.07$ is bracketed by $a = 0$ and $b = 5$. With $C_0 = \binom{29}{2} = 406$ and $D = 7$, Corollary 5.4 gives $\tau_0 = (4 \cdot 232 - 812)/14 = 58/7$, so $T_\ell \geq 9$ on every secant. Theorem 6.1 with $m = 15$ then demands $\Lambda_0 \geq 465\,\varphi_{15}(9) = 465 \cdot \frac{9}{10} = 418.5$, against $\Lambda_0 = 20$. Either route proves $t_2(4,7) \geq 32$.

7.2 The five cases

Table 1 gives the same data for all five parameter sets. In each, k is the smallest integer satisfying the counting bound (1.1). The global column reports the divisibility obstruction in the two-element cases; the local columns report the one-sided bracket bound and the resulting coverage demand, which exceeds Λ_0 by a wide margin in every case.

Proof of Theorem 1.3. For $\mathrm{PG}(4,7)$ see above. For $\mathrm{PG}(4,8)$, $\mathrm{PG}(4,16)$, and $\mathrm{PG}(6,9)$ the admissible set has two elements $s_1 < s_2$, and $(s_2 - s_1)b_{s_2} = k\theta_{d-1} - s_1\theta_d$ is the fraction shown in the global column, which is not an integer; Proposition 5.1 excludes the cap. For $\mathrm{PG}(6,8)$ the admissible set is $\{29,30,31,43,44\}$, so $w_\Pi \in \{27,28,29,41,42\}$ and the local system (5.2) is feasible, but the mean $R/t = 170235/4681 = 36.37$ is bracketed by $a = 29$ and $b = 41$ with no admissible value between them; Corollary 5.4 gives $\tau_0 = 5009/256$, so

Table 1: Complete caps excluded at the counting bound. $\mathcal{S}$ is the admissible set of hyperplane section sizes; the global column gives the integer that the two-size character equations require to be divisible by $s_2 - s_1$; a, b bracket the mean R/t in the one-sided support $\mathcal{S}_+ - 2$; $\beta = \lceil \tau_0 \rceil$ is the resulting lower bound on every T_ℓ; the last column is the coverage demand $N\varphi_m(\beta)$, to be compared with Λ_0.

d	q	k	Λ_0	$\mathcal{S}$	global	a, b	τ_0	β	$N\varphi_m(\beta)$
4	7	31	20	$\{2, 7\}$	6798/5	$0, 5$	58/7	9	418.5
4	8	37	18	$\{3, 7\}$	7602/4	$1, 5$	5/4	2	444
4	16	97	32	$\{4, 9\}$	144173/5	$2, 7$	21/4	6	3990.9
6	8	293	146	$\{29, 30, 31, 43, 44\}$	—	$29, 41$	5009/256	20	40741.0
6	9	387	44	$\{37, 50\}$	3587183/13	$35, 48$	16940/729	24	71703.4

every secant has $T_\ell \geq 20$, and Theorem 6.1 with $N = 42778$, $m = 146$ demands $\Lambda_0 \geq 42778 \cdot \frac{20}{21} > 40740$ against $\Lambda_0 = 146$. In the four two-element cases the one-sided local route of Table 1 gives an independent proof. The arithmetic in the table is replayed by the exact-arithmetic script described below. □

7.3 The scan and where it stops

The tests of Sections 5 and 6 are finite for every (d, q, k). A deterministic script, using only exact integer and rational arithmetic, applies them to every prime power $q \leq 128$ for $d = 3$, $q \leq 64$ for $d = 4$, $q \leq 16$ for $d = 5$, and $q \leq 9$ for $d = 6$, and to every k from the counting bound to twelve above it. For each (d, q, k) it computes Λ_0, $\mathcal{S}$, and the constants (5.1); decides feasibility of the first two equations of (5.2) by a dynamic programme over the gaps of $\mathcal{S} - 2$; solves the global and local systems exactly, with the congruence, when $|\mathcal{S}| \leq 3$; and evaluates the bracket bound, the budget test (6.2), and the ceiling (6.3), in both the two-sided and the one-sided form. Its certificate records, for every (d, q), the counting bound, every excluded k, and the first k surviving all tests.

The outcome in dimension at least four is exactly Theorem 1.3: the five parameter sets above are the only ones in the scanned domain at which the smallest excluded k equals the counting bound, and in each of them $k + 1$ survives every test. (Test firings above the largest cap of a space, for $q = 2$, are not gains and are listed separately in the certificate.) In dimension three the scan reproduces, for most q, the one-step gain of the secant-local bound of [11, Section 5]; in $\mathrm{PG}(3, 7)$ the global divisibility test excludes $k = 12$, where that bound gives 13. The limits of the domain in dimensions five and six are set by the running time of the dynamic programme, not by any mathematical stop. The script, its certificate, and the SHA-256 digest of the certificate are part of the paper's repository; the replay command is in its verification README. None of the exclusions depends on the scan: each is a hand calculation of the kind shown for $\mathrm{PG}(4, 7)$, and the script's role is to replay them and to establish the negative statement about the rest of the domain.

8 Limits of the method

8.1 One step above the counting bound

Passing from k to $k + 1$ increases Λ_0 by $k(q - 1) + 1$, and by Lemma 3.1 each run of the admissible set then has about $\sqrt{2k}$ elements, while the gap between the roots of Q shrinks. In every scanned case the two runs merge at the first step, and once $\mathcal{S}$ is an interval containing the means $k\theta_{d-1}/\theta_d$ and R/t, both integer systems are feasible and every bracket bound is zero. For $\mathrm{PG}(4,7)$ and $k=32$, for example, $\Lambda_0 = 207$ and $Q(s) = \frac{1}{2}(7s^2 - 67s + 192)$ has negative discriminant, so $Q > 0$ everywhere, and $\mathcal{S} = \{0, 1, \ldots, 12\}$. The mechanism therefore gains one step and no more. A larger gain would need a constraint that survives an interval of admissible sizes, and the natural candidate is the distribution of the losses $Q(s_\Pi)$ themselves, which the character equations do not see.

8.2 Complete caps without coplanar quadruples

If $c_4(A) = 0$, every point off A lies on at most one secant. For a complete cap this forces index one everywhere, $\Lambda_0 = 0$, and $Q(s_\Pi) = 0$ for every hyperplane: the section sizes are the integer roots of Q. In the coding dictionary $C_A^\perp$ is then a perfect two-error-correcting code, and Pavese's classification of the equality cases of his counting bound for 4-general sets [7, Proposition 3.1] identifies A as the elliptic quadric of $\mathrm{PG}(3,2)$ or the 11-cap of $\mathrm{PG}(4,3)$. The identity accommodates both: for $k=5$ in $\mathrm{PG}(3,2)$, $Q(s) = (s-1)(s-3)$, and every plane meets the quadric in 1 or 3 points; for $k = 11$ in $\mathrm{PG}(4,3)$, $Q(s) = \frac{3}{2}(s-2)(s-5)$, and every solid meets the cap in 2 or 5 points, the weights 9 and 6 of the dual ternary Golay code. Complete 4-general sets that are not complete caps exist in many other parameters [7, Section 5], and normal rational curves have $c_4 = 0$ in every dimension; the constraints of this paper say nothing about them because they are not complete caps.

8.3 Why degree constraints alone cannot force spread collisions

The higher-dimensional problem of [11] asks whether a positive proportion of the secants of a complete cap in $\mathrm{PG}(d,q)$, $d \geq 4$, must satisfy $T_\ell \geq cq$. The constraints available without an ambient hyperplane count are the incidence identities of Section 2: the secant–point count (2.1), the collision count (2.4), the pencil identities (2.3), the coincidence count $\sum_x \binom{r(x)}{2} = 3c_4(A)$, and the capacity $r(x) \leq m$. We show that formal data can satisfy all of them with almost every secant free.

Proposition 8.1 (Degree constraints do not force spread collisions)**.** *Fix $d \geq 4$ and let $q \to \infty$ with $k = k(q)$ satisfying $k \asymp q^{(d-1)/2}$, $\Lambda_0 \geq 0$, $\Lambda_0 \equiv 0 \pmod 3$, and $\Lambda_0 \leq k(k-1)/24$. Then there exist an abstract k-set A, a family $\mathcal{Q}$ of $M = \Lambda_0/3$ four-subsets of A called coplanar, a finite set Y with $|Y| = \theta_d - k$, and for each pair $\ell \subseteq A$ a $(q-1)$-subset $Y_\ell \subseteq Y$, such that, with $r(x) = \#\{\ell : x \in Y_\ell\}$, $T_\ell = \sum_{x \in Y_\ell}(r(x) - 1)$, and $c_4 = |\mathcal{Q}|$:*

- *(a) $1 \leq r(x) \leq 2 \leq m$ for all $x \in Y$, $\sum_x r(x) = N(q-1)$, and $\sum_x \binom{r(x)}{2} = 3c_4$;*
- *(b) two pairs with disjoint endpoints share a point of Y if and only if their union is in $\mathcal{Q}$, and then they share exactly one point; pairs with a common endpoint share no point;*
- *(c) $T_\ell = \#\{B \in \mathcal{Q} : \ell \subseteq B\}$ and $\sum_\ell T_\ell = 6c_4$, and for every ℓ there are nonnegative integers $z_{\ell,1}, \ldots, z_{\ell,\theta_{d-2}}$ with $\sum_i z_{\ell,i} = k-2$ and $\sum_i \binom{z_{\ell,i}}{2} = T_\ell$;*
- *(d) $T_\ell \in \{0,1\}$ for every ℓ, so at most $6M = 2\Lambda_0 = o(N)$ pairs have $T_\ell > 0$.*

The hypotheses hold for all large q whenever k is within a bounded distance of the smallest integer satisfying the counting bound and $\Lambda_0 \equiv 0 \pmod 3$.

Proof. A maximal family of edge-disjoint four-subsets of A, regarded as copies of K_4 in K_k, has at least $k(k-1)/72$ members: every four-subset shares an edge with some member, and a fixed member shares an edge with at most $6\binom{k-2}{2}$ four-subsets, itself included, so the family has at least $\binom{k}{4}/6\binom{k-2}{2} = k(k-1)/72$ members. Choose $M = \Lambda_0/3 \leq k(k-1)/72$ of them as $\mathcal{Q}$. For each $B = \{a,b,c,d\} \in \mathcal{Q}$ introduce three new points $x_{ab|cd}, x_{ac|bd}, x_{ad|bc}$ and put each into the sets of its two pairs; since the members of $\mathcal{Q}$ are edge-disjoint, each pair lies in at most one member, so each pair receives at most one such point. Complete every Y_ℓ with private points to size $q-1$. The shared points have $r=2$ and there are

$3M$ of them; the private points have $r = 1$ and there are $N(q-1) - 6M$ of them; so $|Y| = N(q-1) - 3M = N(q-1) - \Lambda_0 = \theta_d - k$, and (a) and (b) hold by construction. A pair in a member of $\mathcal{Q}$ has $T_\ell = 1$, all others $T_\ell = 0$, which gives (d) and the first half of (c). For the pencil rows take one entry 2 (for the pair $\{c, d\}$) and $k-4$ entries 1 when $T_\ell = 1$, and $k-2$ entries 1 when $T_\ell = 0$, padded with zeros; this needs $\theta_{d-2} \geq k-2$, which holds for large q because $\theta_{d-2} \geq q^{d-2}$ and $k \asymp q^{(d-1)/2}$ with $d \geq 4$. Finally, near the counting bound $\Lambda_0 = O(kq)$, and $kq = o(k^2)$ since $k \asymp q^{(d-1)/2}$ with $d \geq 4$; so $\Lambda_0 \leq k(k-1)/24$ for large q. $\square$

The data of the proposition are not a cap: no ambient space is present, and the two-point sets Y_ℓ are not lines. Its role is to show that any argument forcing a positive proportion of secants to carry $T_\ell \geq cq$ must use more than the identities (a)–(c). The hyperplane loss identity is such an additional constraint: it ties the distribution of the indices to the hyperplane sections, which the formal data do not have. Whether it can be pushed beyond the one-step gain of Section 7 is the question we leave open.

9 Conclusion and open problems

The counting bound for complete caps has a slack, and the slack has a geometry: every hyperplane carries the prescribed share $Q(s_\Pi)$ of it, so near the counting bound the hyperplane sections of a complete cap have sizes in a set of two or three short runs. The classical character equations, over all hyperplanes and over the hyperplanes through a secant, then become integer systems on a small support, and their infeasibility, or the collision count they force on every secant, excludes the cap. In the coding dictionary the statement is that a quasi-perfect $[k, k-d-1, 4]_q$ code of length equal to the sphere-covering bound has a weight distribution supported on a set that the first two power moments cannot realize. Five parameter sets in dimensions four and six are excluded in this way, and the exact scan shows that in the tested range the mechanism gains exactly one step.

Three questions remain.

(1) *Beyond one step.* One step above the counting bound the admissible set is an interval and every test of this paper is satisfied. The losses $Q(s_\Pi)$ are then still confined to $[0, \Lambda_0]$, and their distribution over the hyperplanes through a secant, rather than the section sizes alone, is the information not yet used. Is there a second-order constraint on that distribution that survives an interval of admissible sizes?

(2) *Higher moments.* The third moment of the section sizes through a secant counts triples of further cap points by the dimension of their span with the secant, and brings in the five-subsets contained in a solid, the weight-five words of $C_A^\perp$. Together with a lower bound on that count it would sharpen Theorem 5.2; no such bound is known to us.

(3) *The proportion problem.* Proposition 8.1 shows that the incidence identities alone cannot force collisions on a positive proportion of secants. Can the hyperplane loss identity, perhaps in the form of Proposition 3.2 applied to sub-configurations, supply the missing constraint?

The scan also raises a small empirical question: dimensions four and six each yield two exclusions in the tested range and dimension five none. The constant $C_0 = \binom{k-2}{2}\theta_{d-4}$ vanishes only for $d \leq 3$, so this is not an artefact of the moment formula; whether it reflects the parity of d or only small numbers would be settled by extending the scan in dimension five.

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